# Explainer: Problem 3 (Rudin 7.8) — Step Function Series

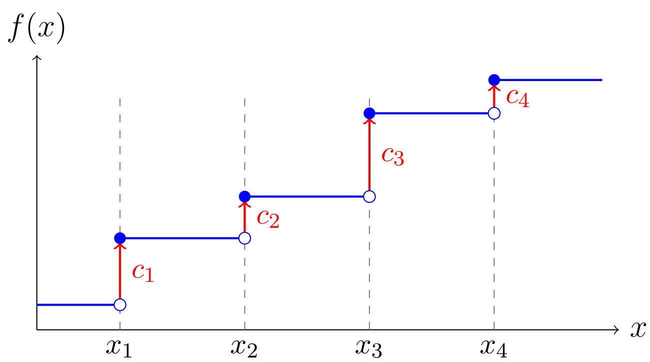
## What's going on?

We build a function by stacking up "jumps." Each jump happens at a point $x_n$ and has size $c_n$:

$$f(x) = \sum_{n=1}^{\infty} c_n \, I(x - x_n), \qquad I(t) = \begin{cases} 0 & t \le 0 \\ 1 & t > 0 \end{cases}$$

The function $I(x - x_n)$ is 0 to the left of $x_n$ and 1 to the right. So $c_n I(x - x_n)$ adds a jump of height $c_n$ at $x_n$.

## Visualizing a partial sum




At each $x_n$, the function jumps by $c_n$. Between jump points, the function is constant (hence continuous).

## Why $\sum |c_n| < \infty$ gives uniform convergence

The tool here is the **Weierstrass $M$-test**. Each term satisfies:

$$|c_n \, I(x - x_n)| \le |c_n| \quad \text{for all } x.$$

Since $\sum |c_n| < \infty$, the $M$-test gives uniform convergence immediately.

**That's it.** The $M$-test is the whole argument for uniform convergence.

## Why $f$ is continuous at $x \neq x_n$

Fix a point $x_0 \neq x_n$ for any $n$. Each individual term $c_n I(x - x_n)$ is continuous at $x_0$ (because $I$ only jumps at $x_n$, and $x_0$ is not any $x_n$). Since we have a uniformly convergent series of continuous functions, Theorem 7.12 tells us the sum $f$ is continuous at $x_0$.

## Why $f$ is discontinuous at each $x_n$ (when $c_n \neq 0$)

At $x = x_n$, the function $I(x - x_n)$ jumps from 0 to 1. The other terms $c_m I(x - x_m)$ for $m \neq n$ are continuous at $x_n$ (since the $x_m$ are distinct). So $f$ has a jump of size $c_n$ at $x_n$.

## Proof outline

1. Apply the Weierstrass $M$-test with $M_n = |c_n|$.
2. Cite Theorem 7.12: uniform limit of continuous functions is continuous.
3. Each term is continuous at $x \neq x_n$, so the sum is too.